

Supplementary Materials for

Preclinical evaluation of antisense oligonucleotide therapy in a mouse model of *HNRNPH2*-related neurodevelopmental disorder

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One Sentence Summary: ASO-mediated *Hnrnp2* knockdown induces *Hnrnp1* upregulation and rescues phenotypes in a mouse model of HNRNPH2-related neurodevelopmental disorder.

Materials and Methods

Antisense oligonucleotide design and screening. Synthesis and purification of all chemically modified oligonucleotides was performed as previously described (1, 2). The 2'-O-methoxyethylribose (MOE) gapmer ASOs were 20 nucleotides long, with a central gap segment comprising ten 2'-deoxynucleotides flanked on the 5' and 3' ends by five 2'-MOE modified nucleotides. All internucleoside linkages were phosphorothioate linkages, except those shown as 'o' in the sequences below, which are phosphodiester. Lyophilized ASOs were formulated in PBS without Ca^{2+} and Mg^{2+} (Gibco, 14190).

As previously described (2), ASOs targeting murine *Hnrmph2* or human *HNRNPH2* were designed at Ionis to the murine and human *hnRNPH2* pre-mRNAs, targeting both exonic and intronic sequences. More than 450 ASOs were designed to the murine *Hnrmph2* pre-mRNA, which were screened in primary mouse embryonic cortical cultures isolated from 16-day pups by adding the ASOs to the culture medium (free uptake). A similar number of ASOs were designed to the human *HNRNPH2* pre-mRNA and screened by free uptake in A431 cells (ATTC). The level of target mRNA reduction was assessed by RT-qPCR and normalized to total RNA level (RiboGreen, Quant-iT). Approximately 40 leads from both the human and mouse screens were evaluated in a 4 point dose-response study to confirm activity and to rank the leads. The top 19 murine *Hnrmph2* ASOs that showed the most efficient knockdown of *Hnrmph2* expression compared to untreated controls were then injected intracerebroventricularly in an 8-week, 700 μg activity and tolerability study in 8–12-week-old female C57BL/6J mice. A dose of 700 μg was selected based on prior work showing that this dose generally achieves greater than 70% inhibition for active ASOs and will identify poorly tolerated ASOs. PBS was used as vehicle control. Mice were observed acutely for any adverse effects as previously described (3, 4) and subsequently weighed and assessed for overall health weekly for 8 weeks. At the end of 8 weeks, animals were sacrificed, tissues collected, and RNA extracted to support analysis of ASO activity and tolerability, measuring astrocyte and microglial activation markers. Screening

data for 3 lead murine *Hnrnp2* ASOs is shown in Table S1. Similar screens were performed for the ASOs targeting human *HNRNP2*. Levels of *Hnrnp1*, *Hnrnp2*, and *Aif1* were assessed by RT-qPCR. TaqMan primer/probe sets were employed with 56-FAM_Zen_3IABkFQ probe chemistry (Integrated DNA Technologies). Forward (F), reverse (R), and probe (P) sequences are as follows. Murine *Hnrnp1* – F: CCTTCGTGCAGTTTGCTTC, R: GATCATAATGAGTTCTAACTTCAGCTC, P: ACCTGTGCCCTATTCTTTCCTTGTGTT; Murine *Hnrnp2* is captured in Table S4; Murine *Gapdh* – F: GGCAAATTCAACGGCACAGT, R: GGGTCTCGCTCCTGGAAGAT, P: AAGGCCGAGAATGGGAAGCTTGTCATC.

The sequences of the selected ASOs are as follows: mouse non-targeted control ASO, 5'-CCTATAGGACTATCCAGGAA-3'; mouse *Hnrnp2* ASO1, 5'-GCTTATACCATTCTCTTAT-3'. Human non-targeted control ASO, 5'-AATGCTTTCATAACTTCCAC-3'; human *HNRNP2* ASO2, 5'-GTACAGCTTTTATATCTGGA-3'.

Animals. WT C57BL/6J mice were used for ASO dose response and longitudinal studies, distribution, and tolerance studies, with a balance of males and females. For all other studies, *Hnrnp2* P209L or R206W mutant knockin mice (5) were used, and male and female cohorts were analyzed separately. Heterozygous mutant females were bred to WT C57BL/6J males to generate heterozygous mutant females (*Hnrnp2*^{P209L/X}), hemizygous mutant males (*Hnrnp2*^{R206W/Y}, *Hnrnp2*^{P209L/Y}), and their WT female (*Hnrnp2*^{X/X}) and male (*Hnrnp2*^{X/Y}) littermates. All experiments were performed on generation F12 or later. Animals were group housed under standard conditions. Experimental protocols were approved by the Institutional Animal Care and Use Committee at St. Jude Children's Research Hospital.

Intracerebral ventricular injection of ASOs. Litters were randomly assigned to treatment groups. Control or *Hnrnp2* ASO was diluted to the appropriate concentration in PBS with 0.0135% Fast Green to aid in verification of the injection location. For neonatal treatment,

postnatal day 2 to 4 pups were cryo-anesthetized for 5-10 minutes. The scalp was cleaned with 70% ethanol and freehand injections were performed using a custom 10 μ l Hamilton syringe with 32-gauge needle (80308, Hamilton). Using the bregma and lambda as anatomical references, 3 μ l control ASO, *Hnrnp2* ASO, or vehicle was injected freehand into the left ventricle. After injection, pups were rewarmed in a petri dish with bedding from the dam's cage until the entire litter was injected. When pups were warm, pink, and capable of spontaneous movement, the litter was sprinkled in dam urine, returned to the home cage, monitored to ensure maternal acceptance, and observed closely for 24 hours post procedure. For juvenile treatment, postnatal day 27 to 29 weanlings were anesthetized with isoflurane by inhalation and their hair shaved with clippers to expose the scalp. After the scalp was cleaned with 70% ethanol, lidocaine/bupivacaine was injected subcutaneously, and a sagittal incision was made to expose the skull. Using the bregma and lambda as anatomical references, 3 μ l control ASO, *Hnrnp2* ASO, or vehicle (0.0135% Fast Green in PBS) was injected freehand into the left ventricle using a custom 10 μ l Hamilton syringe with 32 gauge needle (80308, Hamilton). After injection, the incision was closed with tissue adhesive and mice were moved to a heating pad where they were monitored for recovery from anesthesia and acute post-surgery complications. Once fully awake, mice were returned to their home cage with littermates where they were received subcutaneous meloxicam injections every 24 hours for 3 days and monitored for post-operative complications twice daily for 1 week.

Differentiation and ASO treatment of WT human iPSC-derived neurons. Human male BJFF.6 iPSCs were differentiated into cortical neurons with a two-step protocol (pre-differentiation and maturation) as previously described (6). When iPSCs reached 70-80% confluence, cells were washed twice with DPBS and dissociated with Accutase (STEMCELL Technologies) and collected cells were filtered using a cell strainer (STEMCELL Technologies). Filtered cells were centrifuged at 200 rcf for 5 min at room temperature and pellets were

resuspended with medium containing N2 supplement, non-essential amino acids (NEAA), GlutaMAX Supplement and Y-27632 (STEMCELL Technologies) and 1 µg/ml doxycycline hyclate (Sigma Aldrich). Cell counting was performed using 10 µl and a Countess cell counter (Thermo Fisher). Cells were subplated at 2×10^6 cells/well in 6-well dishes coated with Geltrex (Gibco) in knockout Dulbecco's modified Eagle's medium (KO-DMEM)/F12. The medium was changed daily for 3 days and Y-27632 was removed from day 2. For maturation, pre-differentiated precursor cells were washed, dissociated, counted, and subplated at 25×10^4 cells/ml on dishes coated with 50 µg/ml poly-L-ornithine and 1-2 µg/cm² laminin (Millipore Sigma) in BrainPhys neuronal medium (STEMCELL Technologies) containing N2 (Thermo Fisher Scientific), B-27, 20 ng/ml BDNF (PeproTech), 20 ng/ml GDNF (PeproTech), 500 µg/ml dibutyryl cyclic-AMP (Sigma Aldrich), 200 nM L-ascorbic acid (Sigma Aldrich), 1 µg/ml natural mouse laminin (Thermo Fisher Scientific), 1 µM AraC (Sigma Aldrich), Y-27632 (STEMCELL Technologies) and 1 µg/ml doxycycline hyclate. Y-27632 was removed from day 2. Half-medium was changed every other day. iPSC-derived neurons were treated with *HNRNPH2* ASO or vehicle (DPBS) from day 7 to day 21 of neuronal maturation. The ASO was initially added directly to the media to a final concentration of 10 µM, then at each subsequent half-media change to achieve the same final concentration. RNA isolation, protein isolation, and Western blotting in iPSC-derived neurons were performed as described for HEK293T cells below. ddRT-PCR was performed as described for mouse tissue below.

Cell culture, transfection, Western blot, and digital droplet RT-PCR. HEK293T cells were maintained in Dulbecco's Modified Eagle's Medium (DMEM) supplemented with 10% fetal bovine serum. Cells were transfected with RNAiMAX reagent (Thermo Fisher Scientific, 13778075) according to the manufacturer's protocol, using either ON-TARGETplus Non-targeting Pool siRNA (Dharmacon, D-001810-10-05) or ON-TARGETplus human hnRNPH2 siRNA (Dharmacon, L-013245-02-0005). For plasmid-based transfections, cells were

transfected with HA-minigene and/or GFP-hnRNPH2 using Lipofectamine 3000 (Thermo Fisher Scientific, L3000015) according to the manufacturer's instructions. Twenty-four hours post-transfection, cells were lysed on ice for 30 min in lysis buffer (25 mM Tris-HCl pH 7.5, 150 mM NaCl, 1 mM EDTA, 1% NP-40, 5% glycerol [v/v]) supplemented with 1× Complete Protease Inhibitor Cocktail (Thermo Fisher Scientific, 1862209). An aliquot to 3% of the lysate was used as input. Proteins were resolved by electrophoresis on NuPAGE Novex 4–12% Bis-Tris gels (Invitrogen) and transferred to membranes. Membranes were blocked with Intercept TBS blocking buffer (LI-COR, 927-60001) and incubated with primary antibodies followed by species-specific IRDye secondary antibodies. Imaging was performed on an Odyssey CLx system and analyzed on Image Studio software version 5.5 (LI-COR). Details on antibodies used in western blots are provided in Table S5. For RNA isolation, cells were washed twice with DPBS and processed following the manufacturer's instructions using the RNeasy Mini Kit (74106, Qiagen). RNA was quantified using the Quant-iT RiboGreen RNA Assay Kit (R11490, Thermo Fisher Scientific), or NanoDrop 8000 Spectrophotometer (Thermo Fisher Scientific). ddRT-PCR was performed as described for mouse tissues below. Target and reference genes with assay details are listed in Table S4.

RT-PCR, digital droplet RT-PCR, and Western blot in mouse tissue. Mice were anesthetized by isoflurane inhalation and whole brains were harvested and cut in half sagittally before being flash frozen in liquid nitrogen and stored at -80°C until use. For RNA isolation from mouse brain, tissue was treated with RNAlater-ICE (AM7030, Thermo Fisher Scientific) to prevent degradation, processed using the RNeasy Midi Kit with on-column DNase digestion (75144, Qiagen), and quantified with the Quant-it RiboGreen RNA Assay Kit (R11490, Thermo Fisher Scientific). For gene expression by RT-PCR, 400 ng of RNA from mouse brain was converted to cDNA with the SuperScript IV First-Strand Synthesis System (18091050, Thermo Fisher Scientific) using random hexamer primers. Two µl of cDNA was used for PCR with the

Platinum II Hot-Start PCR Master Mix (14001012, Thermo Fisher Scientific). Products were resolved on a 3 % agarose gel, imaged on the ChemiDoc MP system (Bio-Rad), and densitometry performed with ImageJ. For gene expression analysis by digital droplet PCR, 1-10 ng RNA (depending on the target transcript) was used with a one-step RT-ddPCR Advanced Kit for Probes (1864021, Bio-Rad). Data were analyzed with the QX Manager Software Version 2.0 (Bio-Rad), copies were normalized to those of a reference gene with similar expression level, and expression plotted as a percentage of control. Target and reference genes with assay details are listed in Table S4.

For western blots in mouse, brain tissue was processed for total cell RIPA-soluble fractions as previously published (7), or for cytoplasmic and nuclear fractions with the Subcellular Protein Fractionation Kit for Tissues (87790, Thermo Fisher Scientific) according to the manufacturer's protocol. Protein concentrations were determined by BCA protein assay (A55864, Thermo Fisher Scientific), and 15 µg total cell RIPA-soluble fraction, 100 µg cytoplasmic fraction, or 85 µg nuclear fraction was loaded onto 8% WedgeWell gels (NW00080BOX, Thermo Fisher Scientific). Electrophoresis was performed using the Bolt Bis-Tris mini gel system (Thermo Fisher Scientific) and proteins were transferred to PVDF membranes (IB24001, Thermo Fisher Scientific) using the iBlot 2 dry blotting system (Thermo Fisher Scientific). Membranes were blocked with Intercept TBS blocking buffer (LI-COR, 927-60001) and incubated with primary antibodies followed by species-specific IRDye secondary antibodies. Imaging was performed on an Odyssey CLx system and analyzed on Image Studio software version 5.5 (LI-COR). Details on antibodies used in western blots are provided in Table S5.

RNA-seq data analysis. Total RNA was extracted with an RNeasy Universal Midi kit (Qiagen; 75144) as described above. A sequencing library was prepared with a TruSeq Stranded Total RNA Kit (Illumina) and sequenced with an Illumina HiSeq system with 100-bp read length. Total

stranded RNA sequencing data were processed by the internal AutoMapper pipeline. Raw reads were first trimmed (Trim-Galore v0.60), mapped to mouse (GRCm39) genome assembly (STAR v2.7) and then the gene level values were quantified (RSEM v1.31) based on GENCODE annotation (vM22). We obtained the TPM (transcript per million) counts for genes with TPM greater than one in at least one sample and the gene expression analysis was performed with non-parametric ANOVA using Kruskal-Wallis and Dunn's tests on log-transformed TPM counts between three replicates of each experimental group, implemented in Partek Genomics Suite v7.0 software (Partek). The expression of a gene was considered significantly different if the adjusted (through two-stage linear step-up procedure of Benjamini, Krieger and Yekutieli) (8) P value < 0.05 (equivalent to FDR of 5%) and $\log_2FC > 0.322$. For alternative splicing analysis, the aligned and sorted BAM files after STAR alignment were used for AS analysis using rMATS (v4.0.2) (50). A3SS, A5SS, SE, RI, and MXE events were evaluated. Significant AS events from the two-group differential alternative splicing analysis were identified using the following criteria: average RNA-seq read count ≥ 10 in both sample groups; filter out events with average PSI value < 0.05 or > 0.95 in both sample groups; FDR ≤ 0.05 ; $|\Delta PSI| \geq 0.05$ (9).

Immunofluorescence. Mice were anesthetized by isoflurane inhalation and whole brains harvested, cut in half sagittally, postfixed in 10% neutral buffered formalin (NB0345, StatLab), and embedded in paraffin. Alternatively, mice were anesthetized by isoflurane inhalation and transcardially perfused with 4% paraformaldehyde (J19943, Thermo Fisher Scientific) before whole brains were harvested, cut in half sagittally, cryoprotected in 30% sucrose in PBS, and embedded in Tissue Freezing Medium (TFM-C, GeneralData). To assess the distribution and activity of the *Hnrnp2* ASO in the mouse brain, 4- μ m sagittal paraffin sections were cut on a microtome, deparaffinized and subjected to proteinase K antigen retrieval (S3020, Dako) or

heat-induced epitope retrieval (50-187-82, Thermo Fisher Scientific) depending on the primary antibodies used. Sections were then stained according to standard immunofluorescence protocols. For sections stained with both the rabbit anti-ASO and rabbit anti-HNRNPH2 antibodies, the anti-HNRNPH2 antibody was labeled with CF568 using a Mix-n-Stain antibody labeling kit (92235, Biotium). Details on antibodies used are provided in Table S7.

For ASO distribution across the mouse brain, images were acquired with a Leica SP8 confocal microscope or a Lionheart slide scanner. For ASO distribution into brain cell types, images were acquired with a Leica Stellaris confocal microscope. For *Hnrnph2* downregulation in brain cells containing *Hnrnph2* ASO, images were acquired with a Leica Stellaris confocal microscope or a Leica SP8 confocal microscope.

Histopathology. For histopathological evaluation of *Hnrnph2* ASO toxicity, mice were anesthetized by isoflurane, euthanized, perfused transcardially with 10% neutral buffered formalin and subjected to a standard full necropsy. Tissues were dissected, fixed in 10% neutral buffered formalin, and processed using the HistoCore PEGASUS Tissue Processor (Leica Biosystems). Processed tissues were embedded in paraffin and sectioned at a thickness of 4 μ m using a HistoCore AUTOCUT fully automated rotary microtome (Leica Biosystems). Tissue sections were stained with hematoxylin and eosin (H&E) using the HistoCore SPECTRA ST Stainer (Leica Biosystems). Simplex immunohistochemistry was performed on sequential brain sections to detect CD68 (macrophages), CD163 (peripheral monocyte-derived macrophages), IBA1/AIF1 (microglia/macrophages), and GFAP (astrocytes). Details on epitope retrieval, primary antibodies, staining platform and detection system used are summarized in Table S6. Sections were reviewed by a board-certified veterinary pathologist (AMK). Histopathologic lesions and positive reactivity for each immunohistochemistry assay were semi-quantitatively scored for severity (10). For GFAP and IBA1, positive reactivity above baseline expression in the brain was scored.

Audiogenic seizure susceptibility. Susceptibility to audiogenic seizures was tested as previously described (5). Briefly, mice were exposed to a 120-125 dB fire bell inside a clear acrylic box (30 x 30 x 30 cm) for 60 seconds and the intensity of the response (seizure severity score) was categorized as 0 for no response or slight startle, 1 for wild running, 2 for clonic seizures, 3 for tonic seizures, and 4 for respiratory arrest (11).

Motor function tests. Mice were tested for balance and coordination, muscle strength and gait performance as previously described (5). Rotarod analysis was performed on an accelerating rotarod apparatus (IITC Life Science) using a 2-day protocol consisting of a training day with rotation speed set at 4 rpm for 5 minutes, followed by a testing day with rotation speed set to accelerate from 4 to 40 rpm at 0.1 rpm/s. On test day, latency to fall was recorded for four separate trials per mouse, with a 15-minute rest period between each trial, and the average latency to fall across the 4 trials was calculated. The beam walking test was performed using a 2-day, multibeam (MazeEngineers) protocol (12). Total latency to cross all 3 beams and number of hind paw slips on the 12-mm square beam was analyzed. Grip strength was measured using a grip strength meter (Bioseb) as grams of force for front paws in six repeated measurements. The average grip strength across the 6 trials was calculated. For gait analysis, the front and hind paws of each animal were dipped in water-soluble non-toxic red and blue paint, respectively. The animal was then placed in a 70-cm long tunnel lined on the bottom with Whatman filter paper and allowed to walk through one time. Footprints were scanned and analyzed with Image J for stride length, fore- and hind base width, and overlap (13).

Cognitive function tests. Mice were assessed for anxiety in the elevated plus maze, anxiety and locomotion in the open field test, and spatial learning and memory in the Morris water maze using the ANY-maze automated activity monitoring system and software (v7.2) as previously

described (5). Briefly, in the elevated plus maze (Stoelting), mice were placed in the center facing an open arm away from the investigator and allowed to explore the maze undisturbed for 5 minutes (14). Parameters including total distance traveled, time spent in open arms and time spent in the maze center were recorded and analyzed. The animal's entire area was used to score arm entries and exits, with at least 80% of the animal needed for an arm entry to occur, and mice were considered to be in the center zone if not in any arm. Mice that fell off the maze were excluded from the analysis. For the open field test, mice were placed in the center of a 40 cm x 40 cm open field arena (Stoelting) facing away from the investigator and allowed to explore undisturbed for 20 minutes (15). Parameters including total distance traveled and time in the center zone (defined as the center 24 cm x 24 cm of the maze) were recorded and analyzed. The animal's center was used to score zone entries and exits. The Morris water maze (MazeEngineers) consisted of a cued phase to test the physical ability of the mice to complete the task, followed by a training/spatial learning phase and finally, a spatial memory test phase (16). Parameters including mean speed, time in the thigmotaxis zone (within 10 cm of the pool wall), latency to find the visible platform (cued trials), latency to find the hidden platform (training trials), cumulative distance from the platform (training trials), latency to enter the platform location (test trial), and platform location crosses (test trial) were recorded and analyzed.

Supplementary Figures

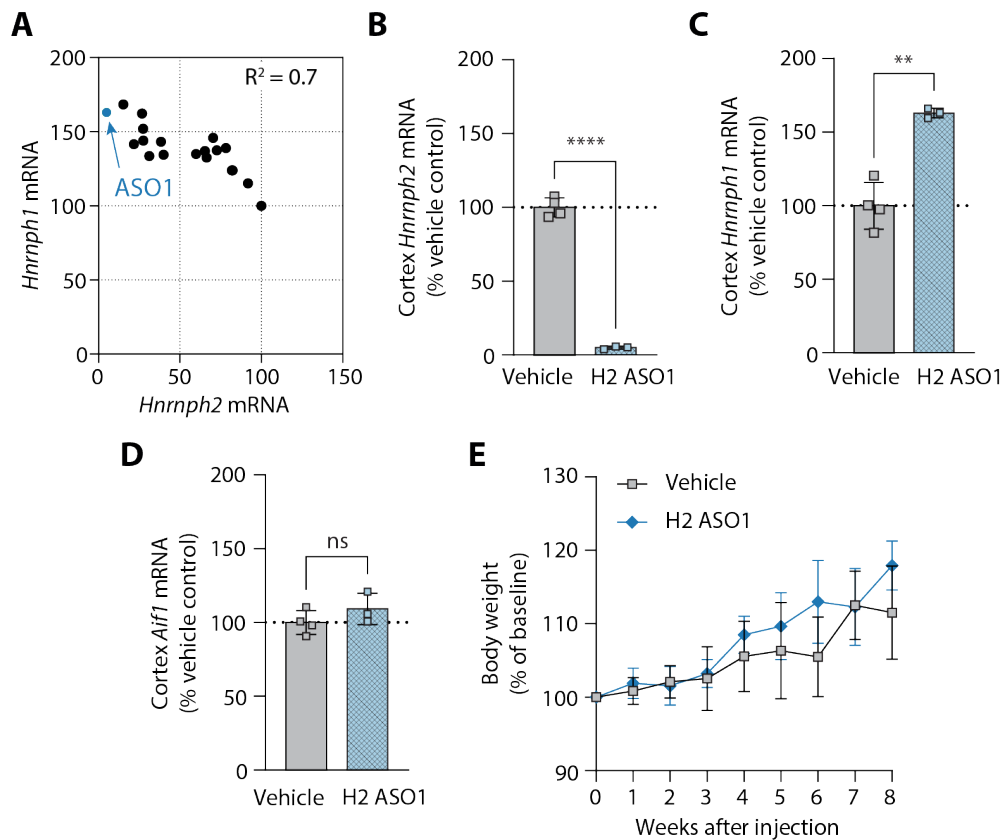


Fig. S1. Initial characterization of mouse *Hnrnp2* ASOs in adult female C57BL/6J mice.

(A) Expression levels of *Hnrnp2* and *Hnrnp1* in mouse cortex 8 weeks after an ICV injection with 700 μ g of the tested *Hnrnp2*-targeted ASOs. Each dot represents a different ASO, with the blue dot indicating ASO1, which was used in all subsequent experiments. mRNA levels were measured by qRT-PCR, normalized to a reference gene, and expressed as a percentage of the vehicle (PBS) control level. There was a significant inverse relationship between *Hnrnp2* and *Hnrnp1* levels ($r = -.8173$, $P < 0.0001$, with an R^2 value of 0.7, $n = 3-4$ mice per group). (B-D) Mouse *Hnrnp2* ASO1 knocks down *Hnrnp2* (B), increases *Hnrnp1* (C), and does not alter expression of microglial activation marker *Aif1* (D) in cortex 8 weeks following a single 700 μ g ICV injection in adult WT mice. mRNA levels were measured by qRT-PCR, normalized to a

reference gene, and expressed as a percentage of the vehicle (PBS) control level. **** $P < 0.0001$, ** $P < 0.01$, unpaired two-tailed t-test, $n = 3-4$ mice per group. Data are represented as mean $\pm$ SD. (E) Body weight was measured weekly across the study. No significant differences were observed in the body weight as a percentage of baseline between the vehicle- and *Hnrmph2* ASO1-treated mice across the study using multiple unpaired t-tests with two-stage step-up (δ) multiple comparisons false discovery rate method.

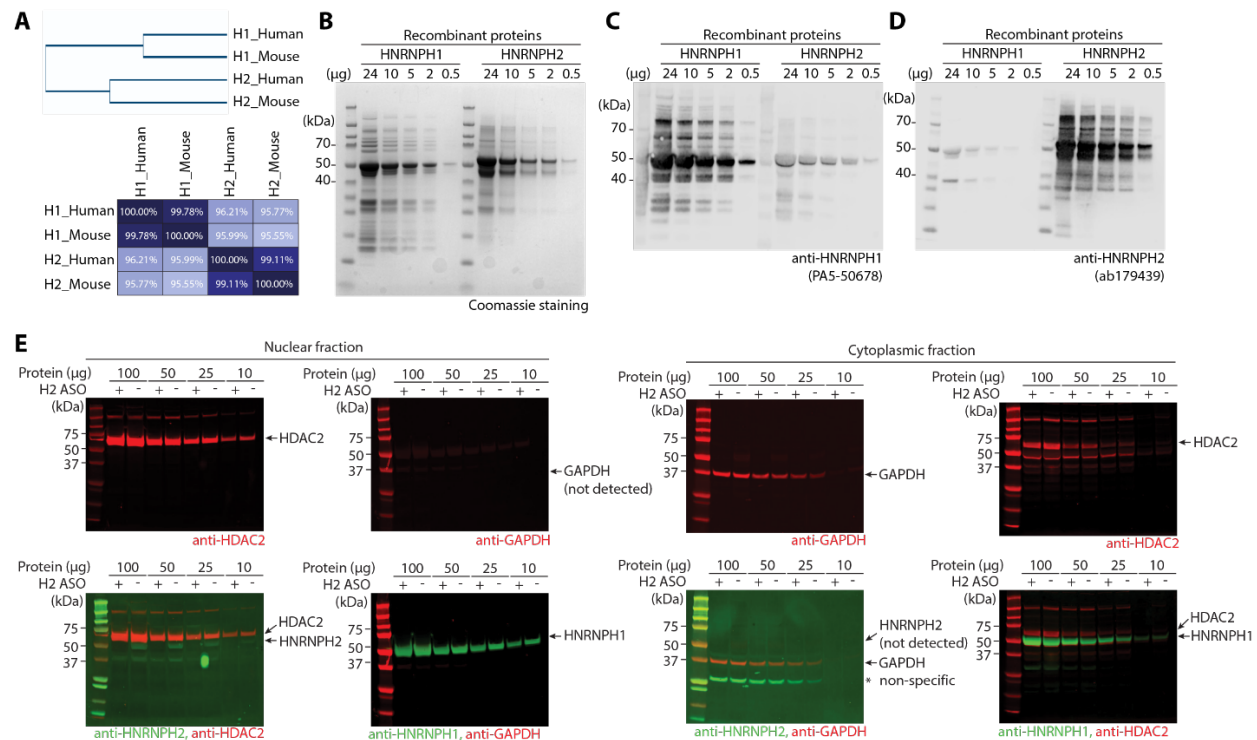


Fig. S2. Validation of antibody specificity and efficiency of subcellular fractionation. (A)

Phylogenetic tree of human and mouse HNRNPH1 and HNRNPH2 proteins. The table displays the percentage identity of their amino acid sequences. **(B-D)** Purified recombinant mouse HNRNPH1 and HNRNPH2 proteins were loaded onto gels in the indicated amounts. The gels were stained with Coomassie blue (B) or blots probed with the indicated antibodies (C, D). **(E)** Nuclear or cytoplasmic fractions prepared from mouse whole brains were loaded onto gels in the indicated amounts. Blots were probed with antibodies against HNRNPH2, a nuclear protein (HDAC2), and a cytoplasmic protein (GAPDH). Nuclear fractions showed a strong HDAC2 signal and no signal for GAPDH, supporting completeness of the fractionation. HNRNPH2 was detected in the nuclear fraction, and the level was reduced in *Hnrmph2* ASO-treated samples. HNRNPH1 was also detected in nuclear fractions. Cytoplasmic fractions showed a strong GAPDH signal as well as some signal for HDAC2, particularly when larger amounts of protein were loaded.

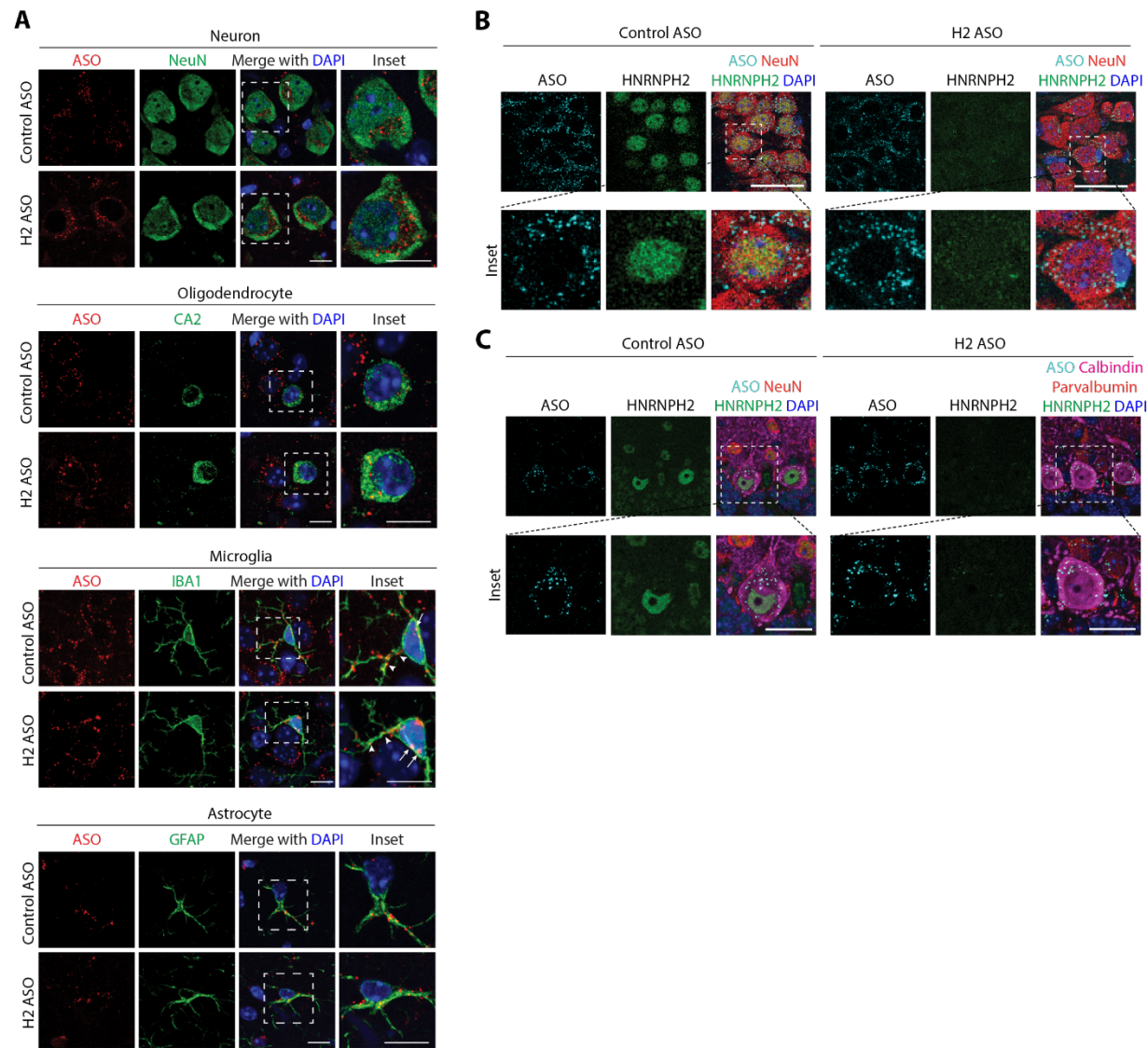


Fig. S3. Distribution and activity of *Hnrnp2* ASO in the mouse brain. WT mice were treated with 30 μ g control or *Hnrnp2* ASO in a single bolus ICV injection at PND 2-4. Brains were harvested at PND 21, processed for immunofluorescence, and slides imaged by confocal microscopy. **(A)** ASO signal was detected inside neocortical NeuN+ neurons, CA2+ oligodendrocytes, and IBA1+ microglia, as well as hippocampal GFAP+ astrocytes in both control ASO- and *Hnrnp2* ASO-treated samples. White arrows and arrowheads indicate ASO present in microglial cell bodies and processes, respectively. **(B)** HNRNPH2 signal was detected in nuclei of NeuN+ neurons in the hippocampus of control ASO-treated brains, but not

in *Hnrnp2* ASO-treated brains. **(C)** HNRNPH2 signal was detected in nuclei of calbindin+ and parvalbumin+ Purkinje cells of the cerebellum in control ASO-treated brains, but not in *Hnrnp2* ASO-treated brains. Scale bars, 7 μ m in (A) and (C), 20 μ M in (B).

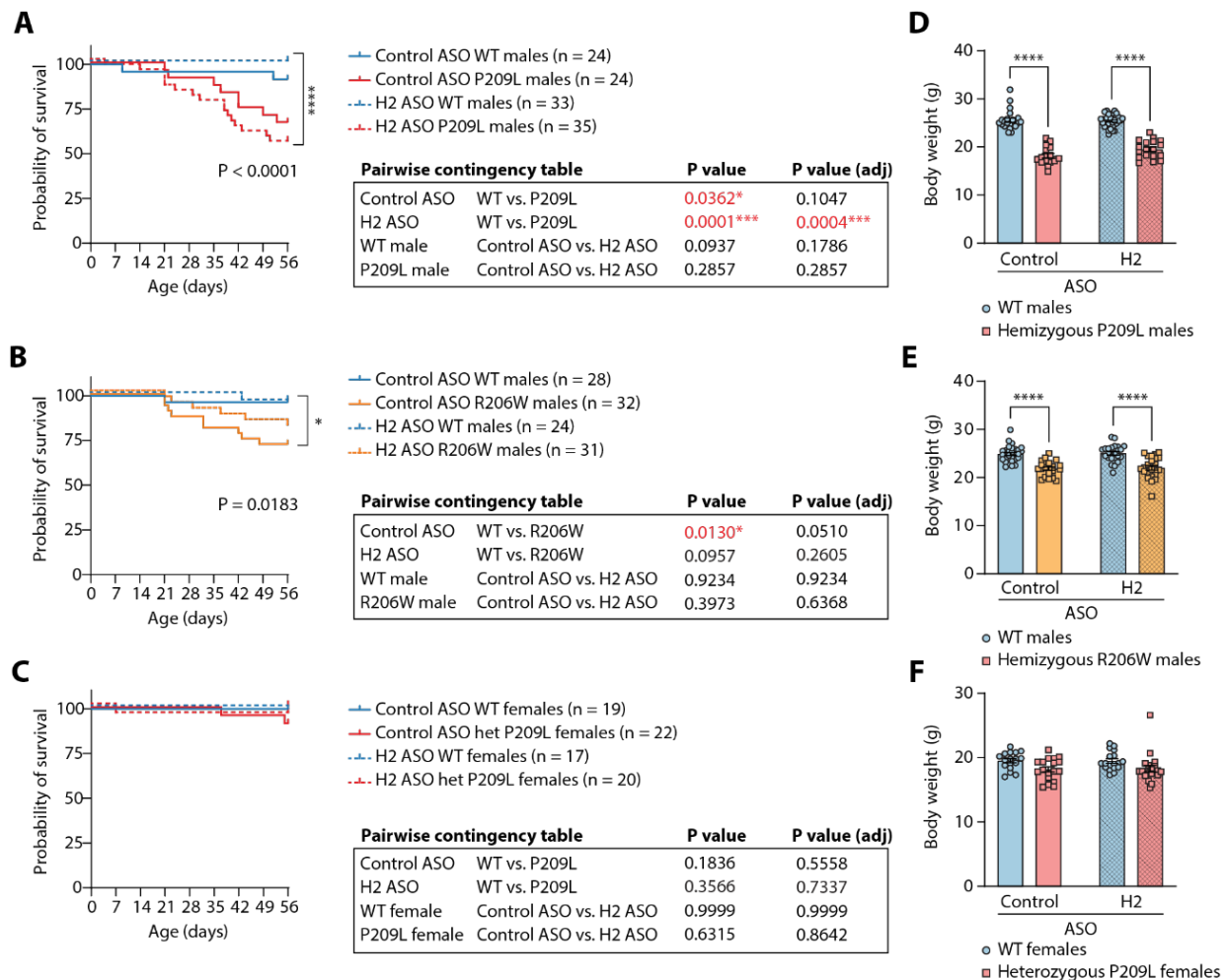


Fig. S4. Neonatal treatment with the *Hnrnp2* ASO does not influence the survival and body weight phenotypes of mutant *Hnrnp2* mice. (A-C) Kaplan-Meier survival curves up to 8 weeks of age for P209L male mice (A), R206W male mice (B), and P209L female mice (C) treated with 30 μ g control or *Hnrnp2* ASO by ICV injection at PND 2-4. **** $P < 0.0001$, * $P = 0.0183$ by Mantel-Cox test. Table shows P values for pairwise survival curve comparisons after adjustment for multiple comparisons by the Holm-Sidak method. (D-F) Effect of neonatal ASO treatment on body weight of P209L male mice (D), R206W male mice (E), and P209L female mice (F) at 8 weeks of age. **** $P < 0.0001$, 2-way ANOVA with Sidak's multiple comparison. For (D): control ASO WT $n = 22$, control ASO P209L $n = 16$, *Hnrnp2* ASO WT $n = 33$, *Hnrnp2*

ASO P209L $n = 18$. For (E): control ASO WT $n = 27$, control ASO R206W $n = 21$, *Hnrnp2* ASO WT $n = 23$, *Hnrnp2* ASO R206W $n = 24$. For (F): control ASO WT $n = 17$, control ASO P209L $n = 18$, *Hnrnp2* ASO WT $n = 17$, *Hnrnp2* ASO P209L $n = 20$. Data are represented as mean $\pm$ SEM.

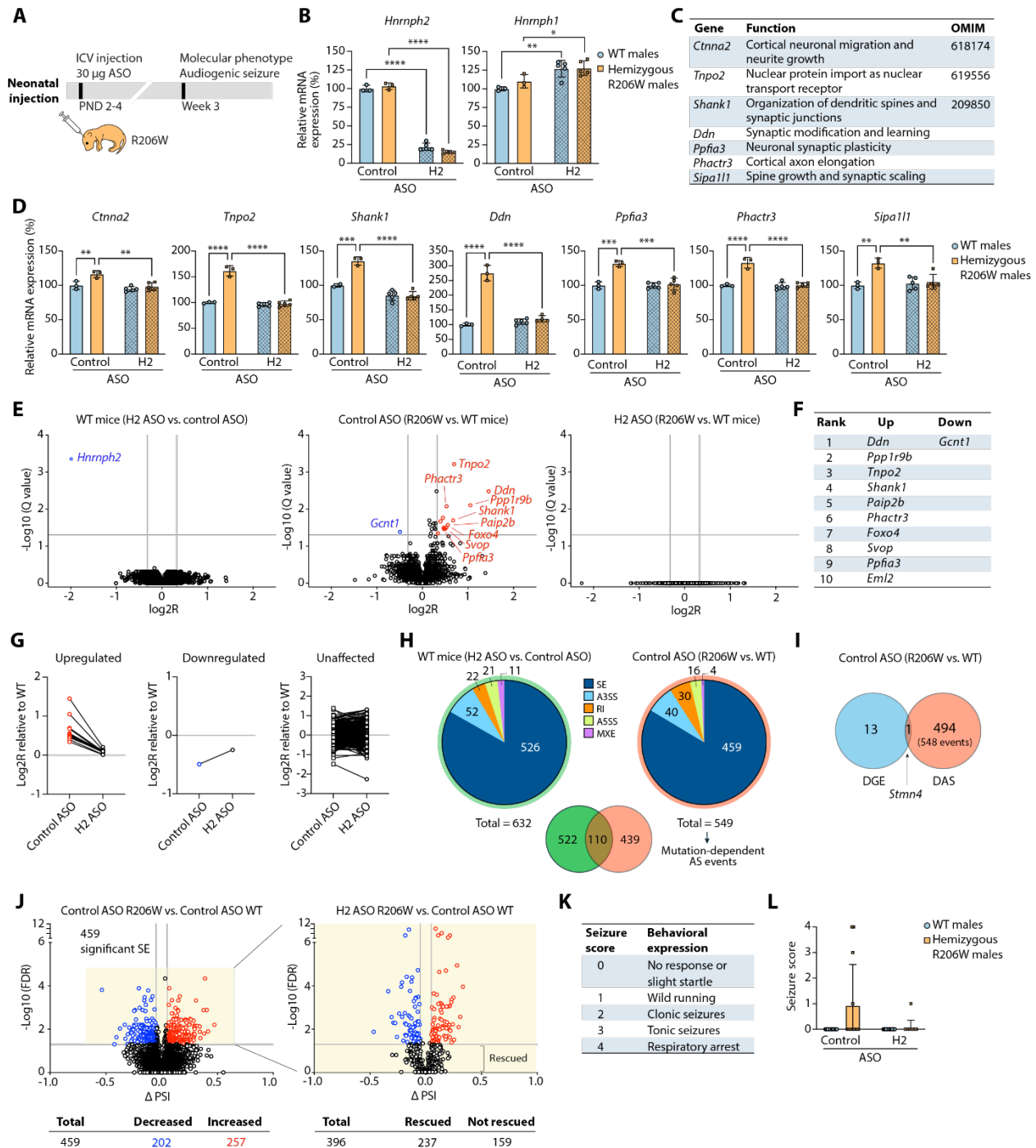


Fig. S5. Neonatal treatment with *Hnrmph2* ASO rescues the molecular phenotype of juvenile R206W male mice. (A) Design of studies used to generate data in (B-L). All panels show data collected from R206W mutant mice or littermates at 3 weeks of age treated with 30 μ g dose of ASO. (B) ddRT-PCR of whole brain tissues showing *Hnrmph2* and *Hnrmph1* mRNA

levels. **(C)** Seven upregulated genes in both *Hnrnp2* mutant mice and human *HNRNPH2* mutant iPSC-derived neurons. **(D)** ddRT-PCR of whole brain tissues showing expression levels of genes listed in (C). **(E)** Volcano plots showing differentially expressed genes with significantly downregulated (blue) and upregulated (red) genes indicated. **(F)** Top 10 most upregulated genes and one downregulated gene by log2R from middle plot in (E). **(G)** Effects of *Hnrnp2* ASO on upregulated (left), downregulated (middle), and unchanged (right) genes at the individual gene level. **(H)** RNA-seq-derived differential alternative splicing events grouped by *Hnrnp2* ASO treatment (left) and R206W mutation (right). SE: skipped exon. A3SS: alternative 3' splice site. RI: retained intron. A5SS: alternative 5' splice site. MXE: mutually exclusive exon. Overlap between *Hnrnp2* ASO-dependent and R206W mutation-dependent events is shown. **(I)** Minimal overlap between R206W mutation-dependent events identified by differential gene expression (DGE) and differential alternative splicing (DAS) analyses. **(J)** Left, volcano plot showing the 459 significant skipped exon (SE) events in control ASO-treated R206W mutant mice with significantly increased (red) and decreased (blue) events. Right, volcano plot showing the effect of *Hnrnp2* ASO treatment on R206W mutation-dependent skipped exon events. **(K-L)** Audiogenic seizure susceptibility scoring (K) and assessment (L) of indicated mice. For panel B and D, 2-way ANOVA with Sidak's multiple comparison, $n = 3-4$ mice per treatment. For panel L, non-parametric Scheirer-Ray-Hare test with Mood's median test, control ASO WT males $n = 11$, control ASO R206W males $n = 13$, *Hnrnp2* ASO WT males $n = 14$, *Hnrnp2* ASO R206W males $n = 13$. For all statistical analyses, **** $P < 0.0001$, *** $P < 0.001$, ** $P < 0.01$, * $P < 0.05$. Data shown as mean $\pm$ SEM.

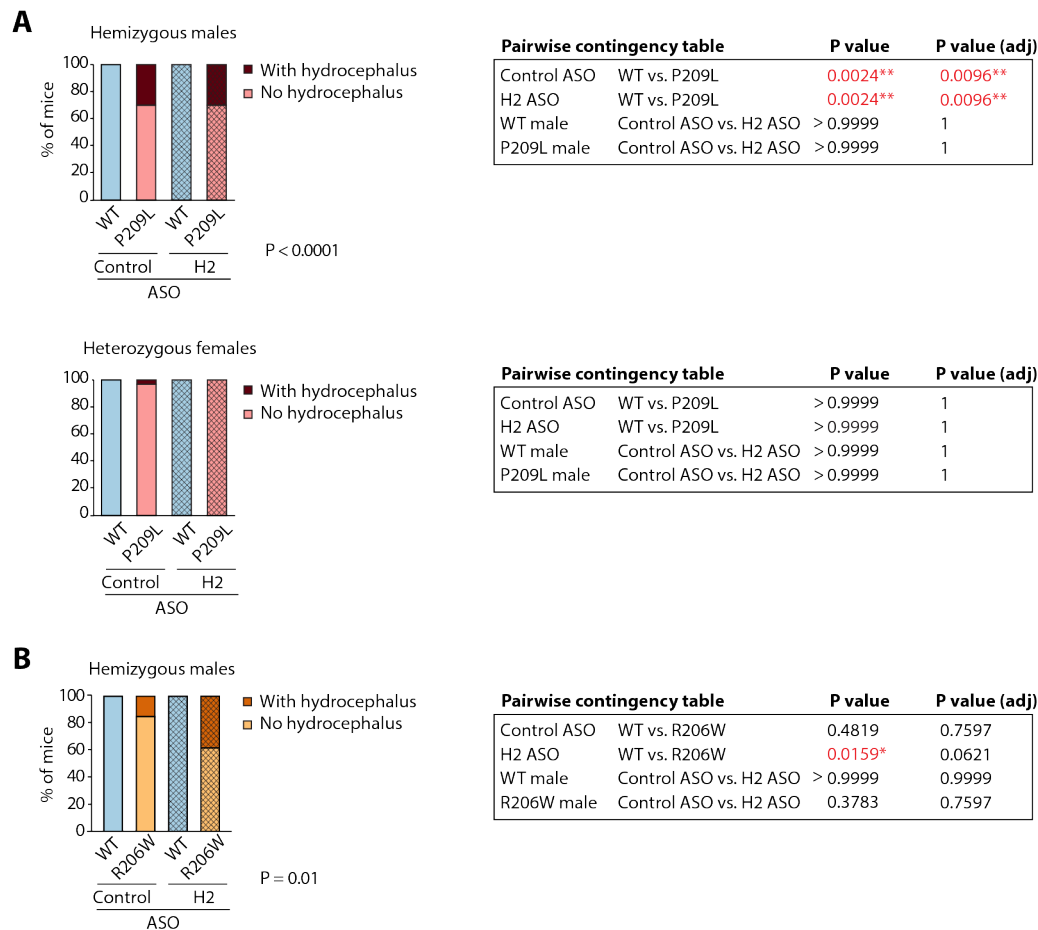


Fig. S6. Effect of neonatal *Hnrnp2* ASO treatment on the incidence of hydrocephalus in male and female mice. Incidence of hydrocephalus observed at dissection of male and female P209L mice (**A**), and male R206W mice (**B**) at 3 weeks of age. $P < 0.0001$, $P = 0.01$, Fisher's exact test. For pairwise comparison, P values were adjusted for multiple comparisons by the Holm-Sidak method.

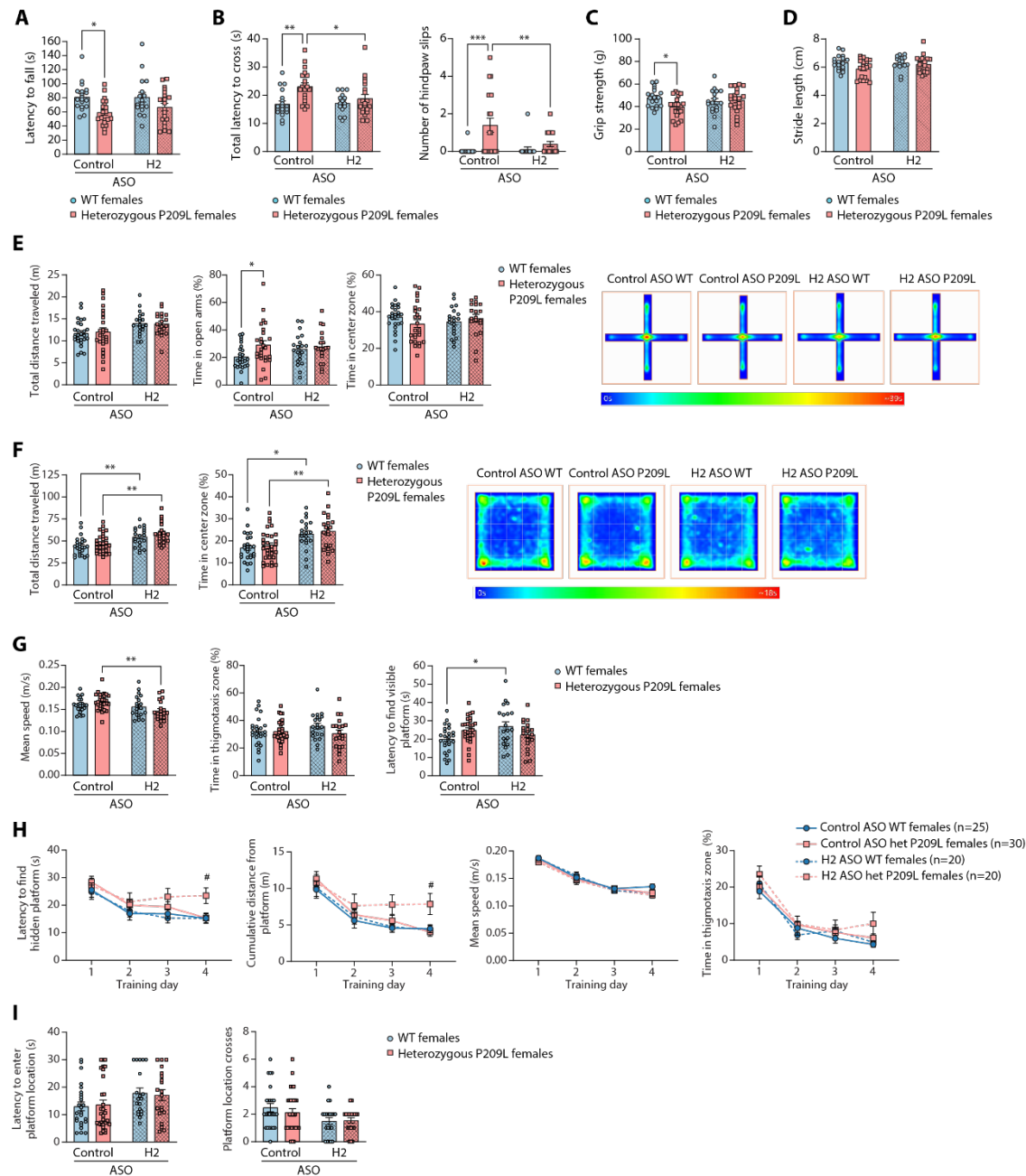


Fig. S7. Impact of neonatal *Hnrnp2* ASO treatment on motor and cognitive functions in adult *Hnrnp2* P209L female mice. (A-D) Effect of neonatal *Hnrnp2* ASO treatment on rotarod performance (A), balance beam performance (B), grip strength (C), and stride length (D) in female P209L mice. (E-I) Effect of neonatal ASO treatment on elevated plus maze behavior (E), open field behavior (F), and Morris water maze performance (G-I) in female P209L mice. In

panel E, heatmaps show the average animals' center point for the groups for the 5-minute test.

In panel F, heatmaps show the average animals' center point for the groups for the 20-minute test. For panels A-G, $**P < 0.01$, $*P < 0.05$ by 2-way ANOVA with Sidak's multiple comparison.

For panel H, $^{\#}P < 0.05$ by linear mixed effects model with random intercept. Post-hoc

comparisons adjusted by FDR within each time point. For panels A-D, control ASO WT females

$n = 19$, control ASO P209L females $n = 20$, *Hnrnp2* ASO WT females $n = 17$, *Hnrnp2* ASO

P209L females $n = 20$. For panel B: control ASO WT females $n = 19$, control ASO P209L

females $n = 20$, *Hnrnp2* ASO WT females $n = 15-17$, *Hnrnp2* ASO P209L females $n = 20$. For

panel E: control ASO WT females $n = 26$, control ASO P209L females $n = 26$, *Hnrnp2* ASO

WT females $n = 20$, *Hnrnp2* ASO P209L females $n = 19$. For panel F: control ASO WT females

$n = 26$, control ASO P209L females $n = 30$, *Hnrnp2* ASO WT females $n = 20$, *Hnrnp2* ASO

P209L females $n = 20$. For panel G-I: control ASO WT females $n = 25$, control ASO P209L

females $n = 30$, *Hnrnp2* ASO WT females $n = 20$, *Hnrnp2* ASO P209L females $n = 20$. Data

are represented as mean $\pm$ SEM.

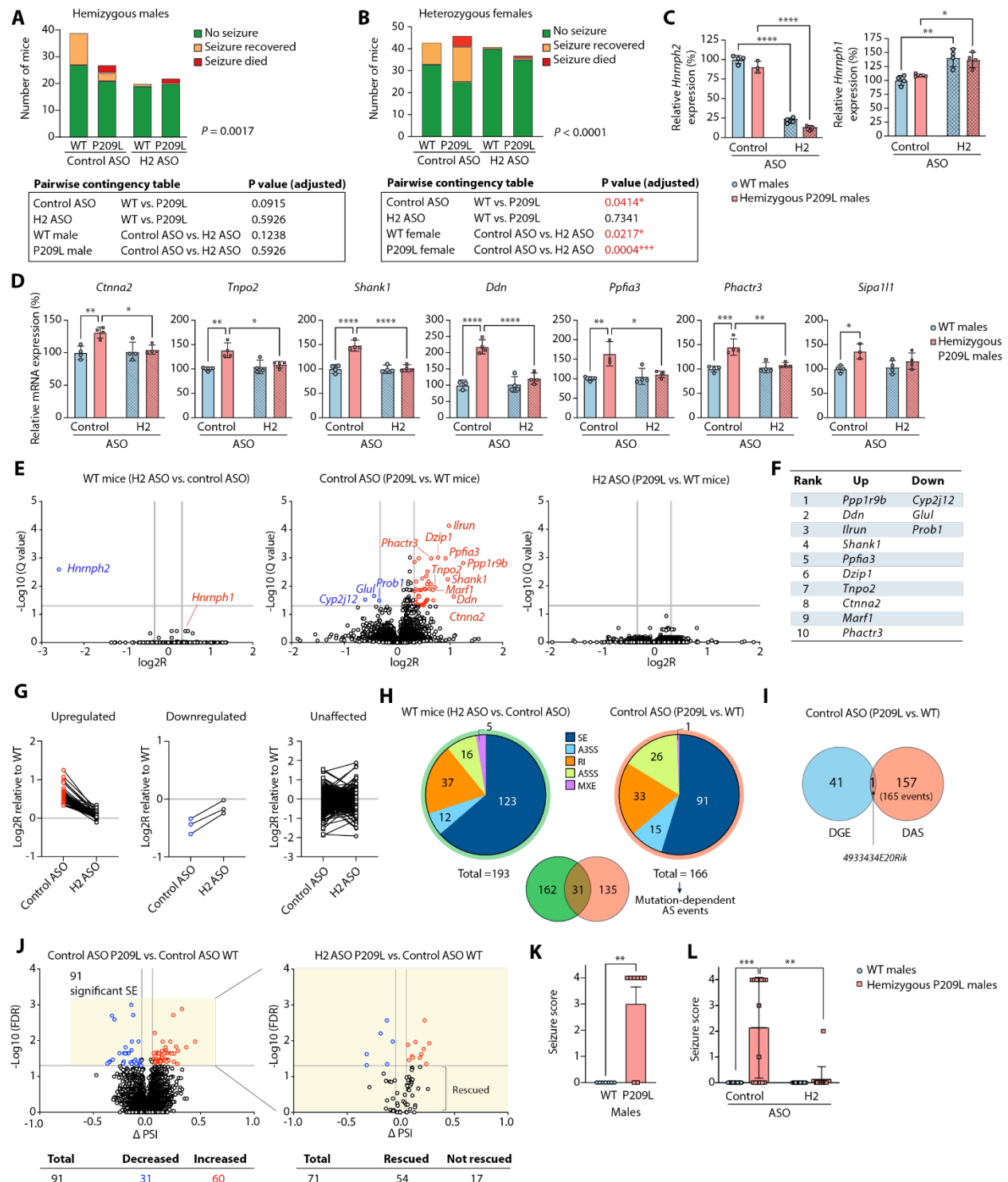


Fig. S8. Juvenile ASO treatment is associated with post-ICV injection seizures and *Hmnp2* ASO rescues the molecular phenotype of juvenile R206W male mice (A-B) Effect

of juvenile ASO treatment on the incidence of post-ICV injection seizures in male (A) and female (B) P209L mice. *P* values next to each graph were determined by Fisher's exact test. Table shows *P* values for pairwise comparisons of contingency tables after adjustment for multiple comparisons by the Holm-Sidak method. For (A), control ASO WT males *n* = 39, control ASO P209L males *n* = 27, *Hnrnp2* ASO WT males *n* = 20, *Hnrnp2* ASO P209L males *n* = 22. For (B), control ASO WT females *n* = 43, control ASO P209L females *n* = 46, *Hnrnp2* ASO WT females *n* = 41, *Hnrnp2* ASO P209L females *n* = 37. (C) ddRT-PCR of whole brain tissues showing *Hnrnp2* and *Hnrnp1* mRNA levels. (D) ddRT-PCR of whole brain tissues showing expression levels of seven genes previously found to be upregulated in both *Hnrnp2* mutant mice and human *HNRNP2* mutant iPSC-derived neurons. (E) Volcano plots showing differentially expressed genes with significantly downregulated (blue) and upregulated (red) genes indicated. (F) Top 10 most upregulated and 3 downregulated genes by log2R from middle plot in (E). (G) Effects of *Hnrnp2* ASO on upregulated (left), downregulated (middle), and unchanged (right) genes at the individual gene level. (H) RNA-seq-derived differential alternative splicing events grouped by *Hnrnp2* ASO treatment (left) and P209L mutation (right). SE: skipped exon. A3SS: alternative 3' splice site. RI: retained intron. A5SS: alternative 5' splice site. MXE: mutually exclusive exon. Overlap between *Hnrnp2* ASO-dependent and P209L mutation-dependent events is shown. (I) Minimal overlap between P209L mutation-dependent events identified by differential gene expression (DGE) and differential alternative splicing (DAS) analyses. (J) Left, volcano plot showing the 91 significant skipped exon (SE) events in control ASO-treated P209L mutant mice with significantly increased (red) and decreased (blue) events. Right, volcano plot showing the effect of *Hnrnp2* ASO treatment on P209L mutation-dependent skipped exon events. (K-L) Audiogenic seizure susceptibility in untreated (K) and ASO-treated (L) male mice. For panels C and D, 2-way ANOVA with Sidak's multiple comparison, *n* = 3-4 mice per group. For panel K, non-parametric Mann-Whitney U test, WT males *n* = 7, P209L males *n* = 8. For panel L, non-parametric Scheirer-Ray-Hare test with Mood's median test,

control ASO WT males $n = 25$, control ASO P209L males $n = 15$, *Hnrnp2* ASO WT males $n = 18$, *Hnrnp2* ASO P209L males $n = 16$. For all statistical analyses, **** $P < 0.0001$, *** $P < 0.001$, ** $P < 0.01$, * $P < 0.05$. Panels C-L show data collected from P209L mutant male mice or littermates at 8 weeks of age treated with 200 μg dose of ASO. Data shown as mean $\pm$ SEM.

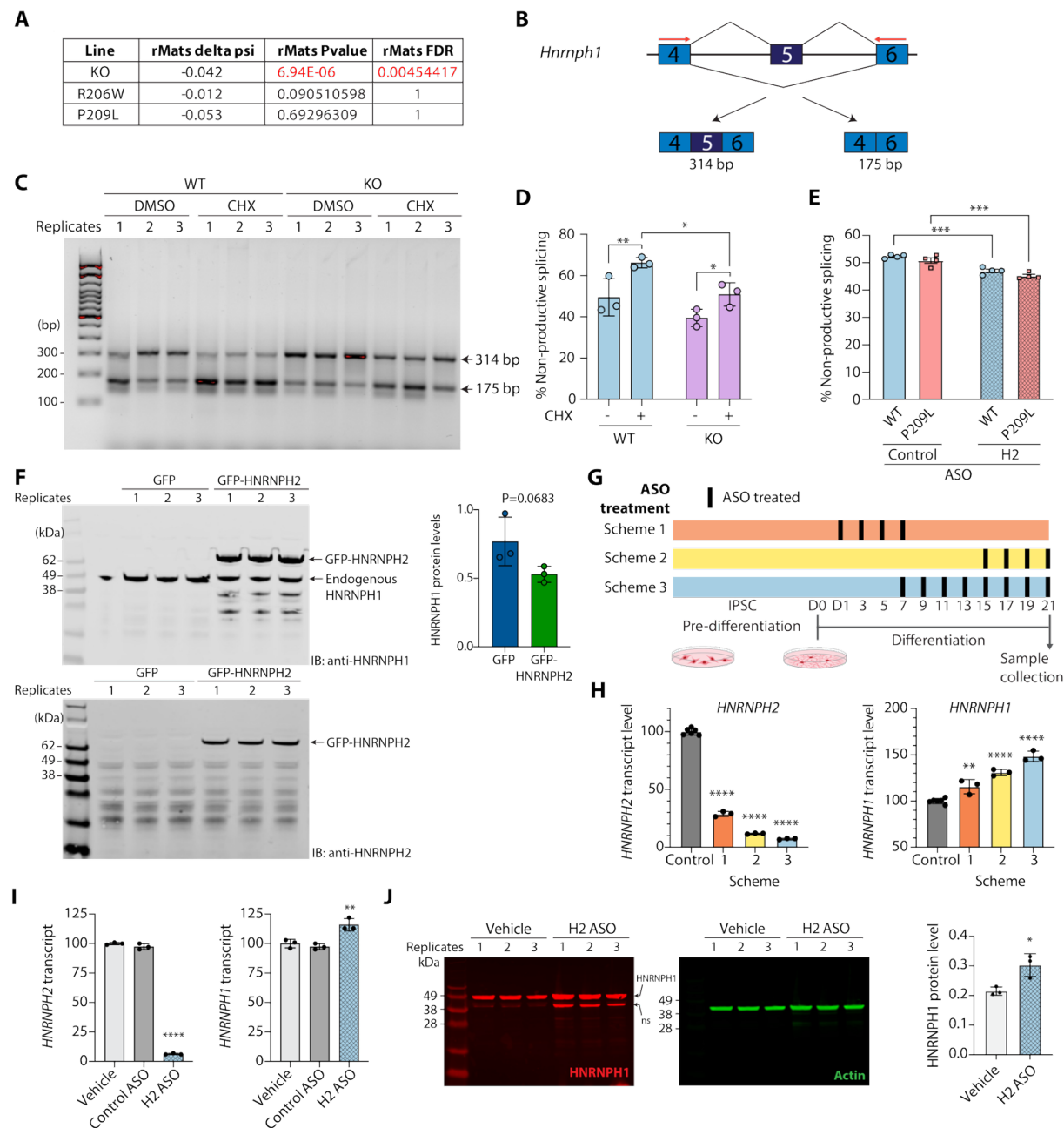


Fig. S9. HNRNP2 protein regulates the expression of *HNRNP1* by modulating its splicing. (A) Significant skipping of essential exon (exon 5) of *Hnrnp1* in cortex from *Hnrnp2* KO mice, but not *Hnrnp2* P209L and R206W mutant mice, as detected by alternative splicing analysis of RNA-seq data (5). (B) Schematic representation of intron-exon structure of murine *Hnrnp1*, including the 139-nt essential exon and flanking exons. Forward and reverse primers were located in exon 4 and 6 (red arrows) and expected sizes of transcripts with and without the

essential exon were 314 bp and 175 bp, respectively. **(C)** Alternative splicing of the essential exon of murine *Hnrnp1* was tested by RT-PCR in the absence (DMSO) or presence of cycloheximide (CHX) in mouse primary cortical neurons. Band identity was confirmed by sequencing. **(D)** Quantification of band intensity in (C) measured by densitometry. The percentage of non-productive splicing, calculated as the level of *Hnrnp1* transcripts lacking the essential exon divided by the level of total *Hnrnp1* transcripts (with and without the essential exon) was significantly increased after cycloheximide treatment in both WT and *Hnrnp2* KO neurons. After cycloheximide treatment, there was a significant reduction in non-productive splicing in *Hnrnp2* KO neurons compared to WT. **(E)** Alternative splicing of the essential exon of murine *Hnrnp1* was reduced by neonatal treatment with 30 µg of *Hnrnp2* ASO in whole brains of juvenile WT and *Hnrnp2* P209L mice. **(F)** Western blot analysis of HEK293T cells transfected with either GFP or GFP-HNRNPH2. Cell lysates were immunoblotted with antibodies against HNRNPH1 (top) and HNRNPH2 (bottom). Note that the HNRNPH1 antibody also recognizes GFP-HNRNPH2, whereas HNRNPH2 antibody only detects overexpressed GFP-HNRNPH2. Endogenous HNRNPH1 protein levels are shown in the accompanying bar graph. **(G)** Schemes for human *HNRNPH2* ASO treatment (10 µg) in human iPSC-derived neurons. **(H)** ddRT-PCR showing levels of *HNRNPH2* and *HNRNPH1* mRNA levels in the three ASO treatment schemes as shown in (G). **(I)** Human iPSC-derived neurons were treated with vehicle, 10 µg of control ASO or human *HNRNPH2* ASO using scheme 3 in G. ddRT-PCR showing levels of *HNRNPH2* and *HNRNPH1* mRNA levels. **(J)** Treatment with human 10 µg *HNRNPH2* ASO in iPSC-derived neurons results in significant increase in HNRNPH1 protein level. Actin was used as a loading control. ns: Non-specific band. **** $P < 0.0001$, *** $P < 0.001$, ** $P < 0.01$, * $P < 0.05$ by 2-way ANOVA with Sidak's multiple comparison (D and E), student's t-test (F and J), one-way ANOVA with Dunnett's multiple comparison (H and I), $n = 3-4$ mice per group (D and E), $n = 3$ technical repeats (F, H-J). Data shown as mean $\pm$ SD.

ASO sequence	Primary screen <i>Hnrmph2</i> mRNA (% control) (20 μ M)	4-point dose-response IC ₅₀ (μ M)	Body weight 8 weeks after dosing (% control)	3 h FOB	Maximum FOB over 8 weeks	<i>Hnrmph2</i> mRNA 8 weeks after dosing - cortex (% control)	<i>Hnrmph2</i> mRNA 8 weeks after dosing – spinal cord (% control)	<i>Aif1</i> mRNA 8 weeks after dosing -spinal cord (% control)	IBA1 IHC 8 weeks after dosing – brain (manual scoring)	CD68 IHC 8 weeks after dosing – brain (manual scoring)
<u>G</u> CTTATACCA TTTCT <u>C</u> TTAT	21.1 $\pm$ 2.0	<1.2	118	0, 0, 0	0, 0, 0	3.8 $\pm$ 0.9	9.0 $\pm$ 0.5	81.6 $\pm$ 2.9	0, 0, 0	0, 0, 0
<u>G</u> TTGTACTTC CCATT <u>T</u> ACCA	29.9 $\pm$ 1.6	<1.2	108	0, 1, 0	0, 0, 0	14.0 $\pm$ 3.4	14.0 $\pm$ 0.1	68.4 $\pm$ 6.3	0, 0, 0	0, 0, 0
<u>T</u> TGTA <u>C</u> TTCC CATT <u>T</u> ACCAA	22.1 $\pm$ 4.2	<1.2	108	0, 0, 0	0, 1, 0	19.0 $\pm$ 1.5	18.0 $\pm$ 2.0	112.4 $\pm$ 20.5	0, 0, 0	0, 0, 0

Table S1. Screening summary of 3 lead mouse *Hnrmph2* ASOs. An example of in vitro and in vivo screening data for 3 lead ASOs designed to target mouse *Hnrmph2*. The ASOs were designed as chimeric 2'-MOE/DNA oligonucleotides with mixed phosphorothioate/phosphodiester linkages. The 2'-MOE nucleotides are underlined. ASOs were initially screened in mouse cortical neurons at 20 μ M concentration. The most active ASOs were evaluated in a 4-point dose-response curve ranging in concentration from 1.2 to 32 μ M. The most active ASOs were evaluated in female C57/BL6 mice (n = 3) at a dose of 700 μ g by ICV injection using a functional observational battery (FOB, scored on a scale of 0 to 7) as described (3, 4). Phosphate-buffered saline was used as vehicle control. Eight weeks after injection, mice were sacrificed (n = 3), spinal cord and cortical tissue were isolated, and RNA was extracted. The relative levels of *Hnrmph2* and *Aif1* mRNA were analyzed using qRT-PCR. Brains were also processed for immunohistochemistry (IHC) for IBA1 and CD68 and scored manually for the presence of neuroinflammation (0 = no different from control).

Tissue & microscopic feature	30 µg control ASO			30 µg <i>Hnrrph2</i> ASO		
	Animal 1	Animal 2	Animal 3	Animal 1	Animal 2	Animal 3
Brain (H&E) Needle tract lesion, cerebral cortex	0 (not present in multiple sections, 3 steps/levels; small focus of hemosiderophages, unilateral, subcallosal, parietal region)	2 (mild, regionally extensive, unilateral, cerebral cortex; additional aggregates of hemosiderophages, leptomeninges, mesencephalic tectum)	0 (not present in multiple sections, 3 steps/levels)	0 (not present in multiple sections, 3 steps/levels)	0 (not present in multiple sections, 3 steps/levels)	0 (not present in multiple sections, 3 steps/levels)
Brain (H&E) Inflammation, mixed immune cell, neuroparenchyma	0	0	0	0	0	0
Brain (H&E) Inflammation, mononuclear immune cell (lymphocyte), neuroparenchyma	0	0	0	0	0	0
Brain (H&E) Necrosis, neuroparenchyma	0	0	0	0	0	0
Brain (H&E) Vacuolation, neuron, cortex, basal nuclei, hippocampus	0	0	0	0	0	0
Brain (H&E, GFAP) Cellularity, increased, astrocyte, neuroparenchyma (excluding needle tracts)	1	1	0	0	0	0
Brain (H&E, CD68) Cellularity, increased, macrophage, neuroparenchyma (excluding needle tracts)	1	1	1	2	1	2
Brain (H&E, CD68, CD163) Cellularity, increased, macrophage, choroid plexus, leptomeninges, perivascular spaces	2	2	2	2	2	2
Brain (H&E, CD68, CD163) Aggregation, vacuolated macrophage, choroid plexus, leptomeninges, perivascular spaces	0	0	0	0	0	0
Brain (H&E, IBA1/AIF1) Cellularity, increased, microglia, neuroparenchyma (excluding needle tracts)	1	2	1	2	1	1
Brain (IHC CD68) [positive score, neuroparenchyma (excluding needle tracts)]	1	1	1	2	1	2
Brain (IHC CD163) [positive score, choroid plexus, leptomeninges, perivascular spaces]	1	1	1	1	1	1
Brain (IHC GFAP) [positive score of GFAP+ cells beyond the extent that would be seen in normal young uninflamed mouse brains]	1	1	0	0	0	0
Brain (IHC IBA1/AIF1) [positive score of IBA1/AIF1+ cells beyond the extent that	1	2	1	2	1	1

would be seen in normal young uninflamed mouse brains]						
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Table S2. Pathology (H&E, IHC) of brain regions. C57BL/6J pups were injected via ICV with a single bolus of 30 µg control ASO, *Hnnrph2* ASO, or vehicle at PND 2-4. At 3 weeks of age, whole brains were harvested and evaluated by histology for signs of neuroinflammation or systemic toxicity. Tissues were evaluated for severity with a scale of 0-5. A score of 0 indicates a negative finding (the lesion/feature (H&E) or positive cells (IHC) are absent in the examined slide). A score of 1 indicates a minimal finding (less than 10% based on a visually estimated approximation). A score of 2 indicates a mild finding (11 to 25% based on a visually estimated approximation). Increasing scores are 3 (moderate), 4 (marked), and 5 (severe) (10). There were no histopathologic (light microscopic) indicators suggestive of systemic toxicity associated with the administration of 30 µg of control or *Hnnrph2* ASO in any of the organs in any of the six animals. There were no noteworthy neuropathologic findings in the brain, spinal cord, dorsal root ganglia, or spinal nerve roots that could be considered related to the administration of 30 µg of control or *Hnnrph2* ASO in any of the six animals. A mild unilateral needle tract lesion was evident in the cerebral cortex in one animal that was administered 30 µg of control ASO. Focal aggregates of GFAP-positive reactive astrocytes, AIF1/IBA1-positive activated microglia, and CD68-positive macrophages were evident within and around this lesion. Another animal that was administered 30 µg of control ASO had a small, focally extensive, unilateral aggregate of GFAP-positive reactive astrocytes, AIF1/IBA1-positive activated microglia, and CD68-positive macrophages suggestive of a needle tract lesion. However, an overt needle tract lesion was not evident in the brain of this animal. The minimal increase in GFAP-positive astrocytes noted in the adjacent brain regions in these two animals was considered related to overt/suspected needle tract lesions. Needle tract lesions were not evident in the brains of the other four animals.

Apart from the lesions noted above, there were no overt neuroinflammatory lesions in any of the brain sections from any of the six animals. However, immunohistochemistry examination revealed minimal to mild increase in AIF1/IBA1-positive activated microglia, and CD68-positive macrophages in the brain in all six animals. Similarly there was minimal to mild increase in the presence of CD68-positive and CD163-positive macrophages in the leptomeninges, choroid plexus, and neuroparenchymal perivascular spaces. These findings were considered related to class-specific, sequence-independent accumulation of macrophages consequent to ASO administration.

However, related class-specific effects that are often reported in some nonclinical safety studies such as intra-lysosomal accumulation of ASO cargo (often independent of sequence specificity) seen light microscopically as the accumulation of

basophilic granules in renal proximal convoluted tubules and the presence of characteristic vacuolated macrophages in multiple tissues were not evident in any of the six animals.

Tissues	30 µg control ASO			30 µg <i>Hnrnp2</i> ASO		
	Animal 1	Animal 2	Animal 3	Animal 1	Animal 2	Animal 3
Dorsal root ganglia	NNATMF	NNATMF	NNATMF	NNATMF	NNATMF	NNATMF
Spinal cord, cervical	NNATMF	NNATMF	NNATMF	NNATMF	NNATMF	NNATMF
Spinal cord, lumbar	NNATMF	NNATMF	NNATMF	NNATMF	NNATMF	NNATMF
Spinal cord, thoracic	NNATMF	NNATMF	NNATMF	NNATMF	NNATMF	NNATMF
Spinal nerve roots	NNATMF	NNATMF	NNATMF	NNATMF	NNATMF	NNATMF
Heart	NNATMF	NNATMF	NNATMF	NNATMF	NNATMF	NNATMF
Muscle, skeletal, appendicular	NNATMF	NNATMF	NNATMF	NNATMF	NNATMF	NNATMF
Trachea	NNATMF	NNATMF	NNATMF	NNATMF	NNATMF	NNATMF
Lung	NNATMF	NNATMF	NNATMF	NNATMF	NNATMF	NNATMF
Gland, thyroid	NNATMF, OPAE	NNATMF, OPAE	NNATMF, OPAE	NNATMF	NNATMF	NNATMF, OPAE
Gland, parathyroid	NPOS/NAE	NPOS/NAE	NPOS/NAE	NPOS/NAE	NPOS/NAE	NNATMF, OPAE
Esophagus	NNATMF	NNATMF	NNATMF	NNATMF	NNATMF	NNATMF
Kidney	NNATMF	NNATMF	NNATMF	NNATMF	NNATMF	NNATMF
Liver	NNATMF	NNATMF	NNATMF	NNATMF	NNATMF	NNATMF
Gall bladder	NPOS/NAE	NNATMF	NNATMF	NNATMF	NNATMF	NNATMF
Gland, adrenal	NNATMF	NNATMF	NNATMF	NNATMF	NNATMF	NNATMF
Thymus	NNATMF	NNATMF	NNATMF	NNATMF	NNATMF	NNATMF
Spleen	NNATMF	NNATMF	NNATMF	NNATMF	NNATMF	NNATMF
Lymph nodes	NNATMF	NNATMF	NNATMF	NNATMF	NNATMF	NNATMF
Gland, mammary	NNATMF	NPOS/NAE	NPOS/NAE	NNATMF	NNATMF	NPOS/NAE
Glands, salivary (parotid, mandibular)	NNATMF	NNATMF	NNATMF	NNATMF	NNATMF	NNATMF
Stomach	NNATMF	NNATMF	NNATMF	NNATMF	NNATMF	NNATMF
Pancreas	NNATMF	NNATMF	NNATMF	NNATMF	NNATMF	NNATMF
Small intestines (duodenum, jejunum, ileum)	NNATMF	NNATMF	NNATMF	NNATMF	NNATMF	NNATMF
Gut associated lymphoid tissue (GALT)	NNATMF	NNATMF	NNATMF	NNATMF	NNATMF	NNATMF
Large intestines (cecum, colon, rectum)	NNATMF	NNATMF	NNATMF	NNATMF	NNATMF	NNATMF
Reproductive tract, ovary	NNATMF, OPAE	NNATMF	[male animal]	NNATMF	NNATMF	NNATMF
Reproductive tract, oviduct	NNATMF, OPAE	NNATMF	[male animal]	NNATMF	NNATMF	NNATMF
Reproductive tract, uterus/cervix	NNATMF	NNATMF	[male animal]	NNATMF	NNATMF	NNATMF
Reproductive tract, vagina	NNATMF	NNATMF	[male animal]	NNATMF	NNATMF	NNATMF
Testes	[female animal]	[female animal]	NNATMF immature (no spermatogenesis yet)	[female animal]	[female animal]	[female animal]
Epididymis	[female animal]	[female animal]	NNATMF	[female animal]	[female animal]	[female animal]

Gland, prostate	[female animal]	[female animal]	NNATMF	[female animal]	[female animal]	[female animal]
Seminal vesicles	[female animal]	[female animal]	NPOS/NAE	[female animal]	[female animal]	[female animal]
Urinary bladder	NNATMF	NPOS/NAE	NNATMF	NNATMF	NPOS/NAE	NNATMF
Skin	NNATMF	NNATMF	NNATMF	NNATMF	NNATMF	NNATMF
Eyes	NNATMF	NNATMF, ITAE2	NNATMF	NNATMF	NNATMF	NNATMF
Gland, Harderian	NNATMF	NNATMF	NNATMF	NNATMF	NNATMF	NNATMF
Gland, pituitary	NNATMF	NNATMF	NPOS/NAE	NPOS/NAE	NPOS/NAE	NNATMF
Head, cranial nerves and ganglia	NNATMF	NNATMF	NNATMF	NNATMF	NNATMF	NNATMF
Head, ear	NNATMF	NNATMF	NNATMF	NNATMF	NNATMF	NNATMF
Head, nasal cavity	NNATMF	NNATMF	NNATMF	NNATMF	NNATMF	NNATMF
Head, oral cavity	NNATMF	NNATMF	NNATMF	NNATMF	NNATMF	NNATMF
Head, teeth	NNATMF	NNATMF	NNATMF	NNATMF	NNATMF	NNATMF
Head, tongue	NNATMF	NNATMF	NNATMF	NNATMF	NNATMF	NNATMF
Nerve, optic	NNATMF	NNATMF	NNATMF	NNATMF	NNATMF	NNATMF
Muscle, skeletal, axial	NNATMF	NNATMF	NNATMF	NNATMF	NNATMF	NNATMF
Sternum	NNATMF	NNATMF	NNATMF	NNATMF	NNATMF	NNATMF
Bone marrow (H&E)	NNATMF	NNATMF	NNATMF	NNATMF	NNATMF	NNATMF
Femur	NNATMF	NNATMF	NNATMF	NNATMF	NNATMF	NNATMF
Joint, femorotibial	NNATMF	NNATMF	NNATMF	NNATMF	NNATMF	NNATMF

Table S3. Pathology (H&E, IHC) of all organs. C57BL/6J pups were injected via ICV with a single bolus of 30 µg control ASO, *Hnrnp2* ASO, or vehicle at PND 2-4. At 3 weeks of age, mice were subjected to a standard full necropsy and tissues evaluated by histology for signs of neuroinflammation or systemic toxicity. ASEN = All structures of the middle and inner ear are not available for evaluation in every section/every animal; limited tissue available for examination; ITAE = Insufficient tissue available for evaluation; ITAEC = Insufficient tissue available for evaluation in the second/contralateral eye; NNATMF = No noteworthy ASO-related toxicologically relevant microscopic findings; NPOS/NAE = Not present on the slide/not available for examination; OPAE = one of a pair available for evaluation.

Table S4. Gene expression assays used for mouse brain ddRT-PCR and RT-PCR, and human ddRT-PCR.

Gene	Target/Reference	Fluorophore	Assay ID/custom primer probe sequence	Supplier
<i>Hnrnp1</i>	Target	FAM	Mm00517601_m1	Thermo Fisher Scientific
<i>Hnrnp1</i> with essential exon	Target	FAM	Probe: TGGCTTTGTACGGCTTAGAGGACT Forward primer: ATAGTCCTGACACAGCCAATG Reverse primer: CGATTTCCAACCCTGAGAAGA	IDT
<i>Hnrnp1</i> lacking essential exon	Target	FAM	Probe: TGGCTTTGTACGGCTTAGAGGACT Forward primer: TCCTGACACAGCCAATGA Reverse primer: TCAATATACCTGAGAAGAACTGAAC	IDT
<i>Hnrnp2</i>	Target	FAM	Mm01340844_g1	Thermo Fisher Scientific
<i>Gfap</i>	Target	FAM	Mm01253033_m1	Thermo Fisher Scientific
<i>Aif1</i>	Target	FAM	Mm00479862_g1	Thermo Fisher Scientific
<i>Cd68</i>	Target	FAM	Mm03047343_m1	Thermo Fisher Scientific
<i>Ddn</i>	Target	FAM	Mm02020418_s1	Thermo Fisher Scientific
<i>Ppfia3</i>	Target	FAM	Mm01203570_g1	Thermo Fisher Scientific
<i>Tnpo2</i>	Target	FAM	Mm00520397_m1	Thermo Fisher Scientific
<i>Ctnna2</i>	Target	FAM	Mm00515399_m1	Thermo Fisher Scientific
<i>Phactr3</i>	Target	FAM	Mm01212666_m1	Thermo Fisher Scientific
<i>Shank1</i>	Target	FAM	Mm01206737_m1	Thermo Fisher Scientific
<i>Sipa111</i>	Target	FAM	Mm01242798_m1	Thermo Fisher Scientific
<i>Rpp30</i>	Reference	HEX	dMmuCPE5097025	Bio-Rad
<i>Tfr</i>	Reference	VIC	Mm00441941_m1	Thermo Fisher Scientific
<i>Pgk1</i>	Reference	VIC	Mm01225301_m1	Thermo Fisher Scientific
<i>Hnrnp1</i>	Target	NA	Forward primer: TGGGTGTTGAAGCATACTGG Reverse primer: CGTGGTGGGTCATAATGAGTT	IDT
HNRNPH1	Target	FAM	Hs01033845_g1	Thermo Fisher Scientific

HNRNPH1 with essential exon	Target	FAM	Probe: TTCTTCTCAGGGTTGAAAT Forward primer: CGGCTTAGAGGACTTCC Reverse primer: CACCGGCAATGTTATCC	IDT
HNRNPH1 lacking essential exon	Target	FAM	Probe: CAGTTCTTCTCAGGTATATTGA Forward primer: CGGCTTAGAGGACTTCC Reverse primer: AGCTTTCGTGGTGGATC	IDT
HNRNPH2	Target	FAM	Hs01395062_m1	Thermo Fisher Scientific
RPP30	Target	HEX	qHsaCEP0052683	Bio-Rad
SDHA	Target	VIC	Hs07291714_mH	Thermo Fisher Scientific

Table S5. Antibodies used in western blots

Antibody	Specificity	Catalog number	Supplier	Host	Dilution used
Anti-HNRPH2 antibody [EPR12170(B)]	Specific for HNRNPH2 (mouse only, detects both human HNRNPH2 and HNRNPH1)	ab179439	Abcam	Rabbit	1:500
hnRNP H1 polyclonal Antibody	Specific for HNRNPH1 (mouse and human)	PA5-50678	Thermo Fisher Scientific	Rabbit	1:1000
GAPDH (D4C6R) Mouse mAb	GAPDH (cytoplasmic protein)	#97166	Cell Signaling Technology	Mouse	1:2000
HDAC2 Monoclonal Antibody (HDAC2-62)	HDAC2 (nuclear protein)	MA1-25451	Thermo Fisher Scientific	Mouse	1:1000
β -Actin (13E5) Rabbit mAb	Total β -actin	4970	Cell Signaling Technology	Rabbit	1:1000
HA tag antibody	HA tag	NB600-363	Novus Bio	Rabbit	1:1000
Anti-GFP	GFP tag	11814460001	Millipore Sigma	Mouse	1:1000
Secondary ab's for human WBs					
IRDye 800CW Donkey anti-Rabbit IgG Secondary Antibody	Heavy & light chains of rabbit IgG, and light chains common to most rabbit immunoglobulins	926-32213	LICOR	Donkey	1:10,000
IRDye 680RD Goat anti-Mouse IgG Secondary Antibody	Heavy & light chains of mouse IgG ₁ , IgG _{2a} , IgG _{2b} , and IgG ₃ , and light chains of mouse IgM and IgA	926-68070	LICOR	Goat	1:10,000

Table S6. Epitope retrieval, primary antibodies, staining platform, and detection system used in histopathology.

Marker	Primary antibody, dilution, and incubation time	Staining platform, antigen retrieval, detection system, and chromogens
CD68	97778, Cell Signaling Rabbit monoclonal antibody (E307V) 1:900, 15 minutes	Leica Bond Rx Heat induced epitope retrieval, Epitope Retrieval Solution 2 (Leica, AR9640), 20 minutes; Bond Polymer Refine Detection Kit, DAB Refine Kit, and Hematoxylin Refine Kit (Leica, DS9800)
CD163	ab182422, Abcam Rabbit monoclonal antibody (EPR19518) 1:1200, 15 minutes	
IBA1 (AIF1)	CP290A, Biocare Medical Rabbit polyclonal antibody 1:300, 32 minutes	Ventana DISCOVERY ULTRA Heat induced epitope retrieval, Cell Conditioning Solution 1 (Ventana, Roche, 950-500), 32 minutes; OmniMap anti-Rabbit HRP Kit (Ventana, Roche, 760-4311); ChromoMap DAB (Ventana, Roche, 760-159); Hematoxylin II (Ventana, Roche, 760-2208); Blueing reagent (Ventana, Roche, 760-2037)
GFAP	Z0334, Agilent Dako Rabbit polyclonal antibody 1:4000, 32 minutes	

Table S7. Antibodies used for mouse brain immunofluorescence

Antibody	Host	Specificity	Dilution used	Catalog number	Supplier
Anti-ASO antibody	Rabbit	ASO phosphorothioate backbone	1:20,000	NA	Ionis
Anti-HNRPH2/HNRNPH2 antibody [EPR12171]	Rabbit	Specific for HNRNPH2	1:100	ab181171	Abcam
Anti-GFAP antibody	Rabbit	GFAP (astrocyte marker)	1:1000	Z033429	Agilent
Anti-GFAP monoclonal antibody (2.2B10)	Rat	GFAP (astrocyte marker)	1:1000	13-0300	Thermo Fisher Scientific
Anti-NeuN antibody	Guinea pig	NeuN (neuronal marker)	1:250	266004	Synaptic Systems
Anti-IBA1 antibody	Guinea pig	IBA1 (microglial marker)	1:500-1000	234308	Synaptic Systems
Anti-Carbonic Anhydrase II antibody	Sheep	Ca2 (oligodendrocyte marker)	1:500	AHP206	Bio-Rad
Anti-Calbindin D-28K antibody	Chicken	Calbindin (Purkinje cell marker)	1:1000	NBP2-50028	Novus Biologicals
Anti-Parvalbumin antibody	Guinea pig	Parvalbumin (Purkinje cell / interneuron marker)	1:500	GP72	Swant
Alexa Fluor 488 anti-rabbit antibody	Donkey	IgG	1:500	711-547-003	Jackson ImmunoResearch
Alexa Fluor 488 anti-rabbit antibody	Donkey	IgG (Fab)	1:500	711-545-152	Jackson ImmunoResearch
Alexa Fluor 568 anti-guinea pig antibody	Donkey	IgG	1:500	A11075	Thermo Fisher Scientific
Alexa Fluor 568 anti-rat antibody	Donkey	IgG (Fab)	1:500	ab175708	Abcam
Alexa Fluor 594 anti-sheep antibody	Donkey	IgG	1:500	713-585-003	Jackson ImmunoResearch
Alexa Fluor 647 anti-chicken antibody	Donkey	IgY	1:500	703-605-155	Jackson ImmunoResearch
Alexa Fluor 647 anti-guinea pig antibody	Donkey	IgG	1:500	706-605-148	Jackson ImmunoResearch
Alexa Fluor 647 anti-rat antibody	Donkey	IgG	1:500	712-605-153	Jackson ImmunoResearch
Alexa Fluor 750 anti-guinea pig antibody	Goat	IgG	1:500	ab175745	Abcam

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